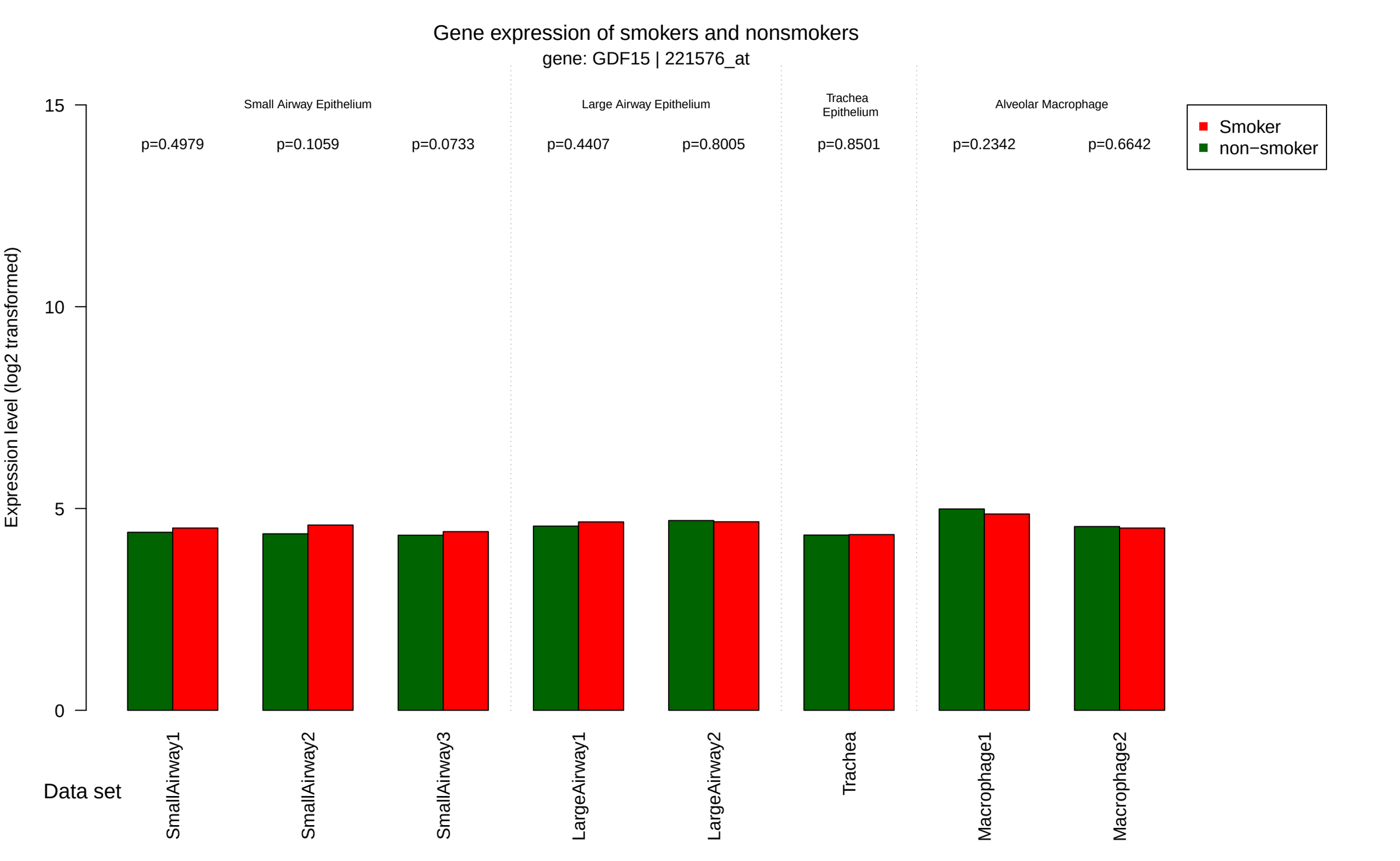
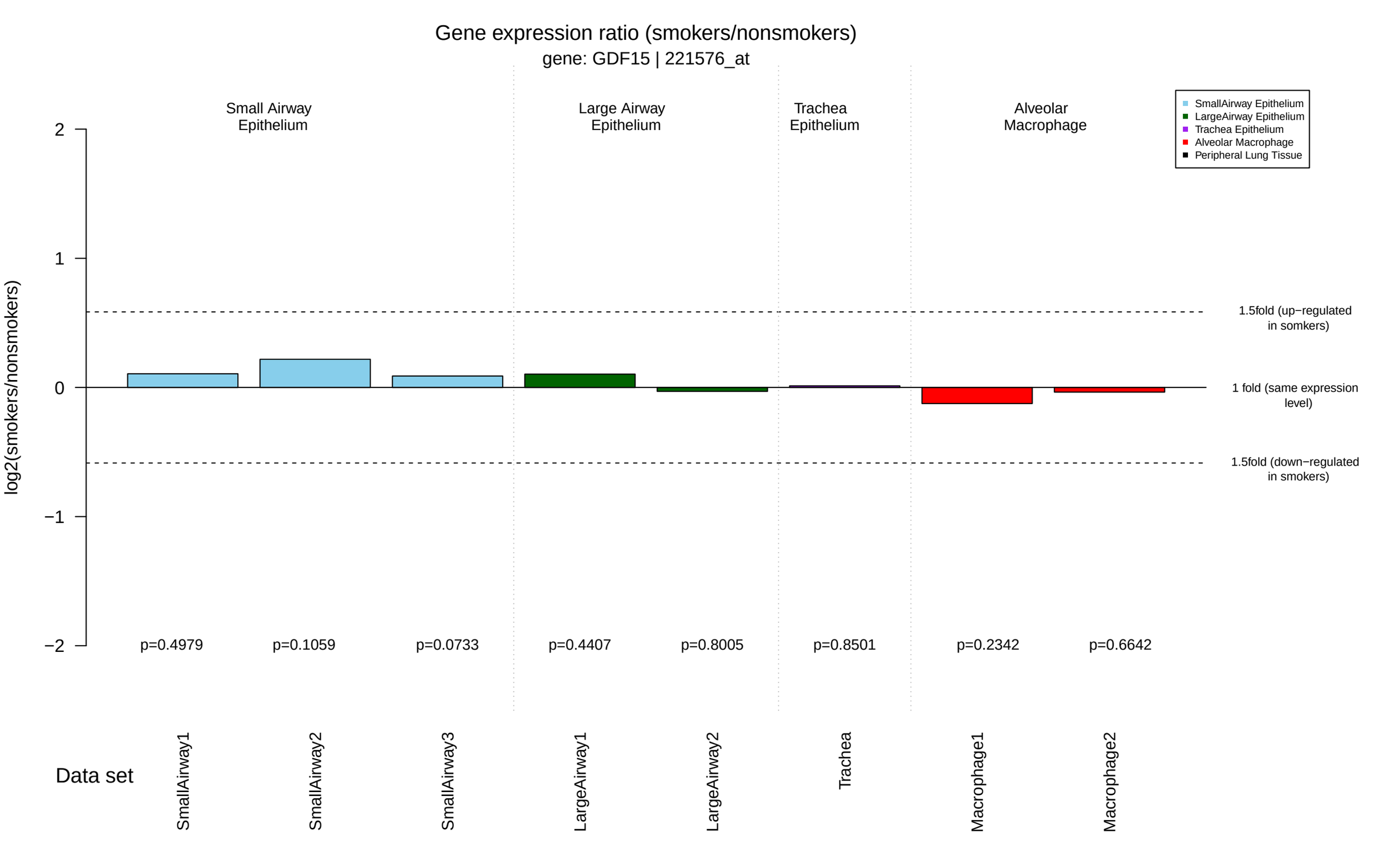
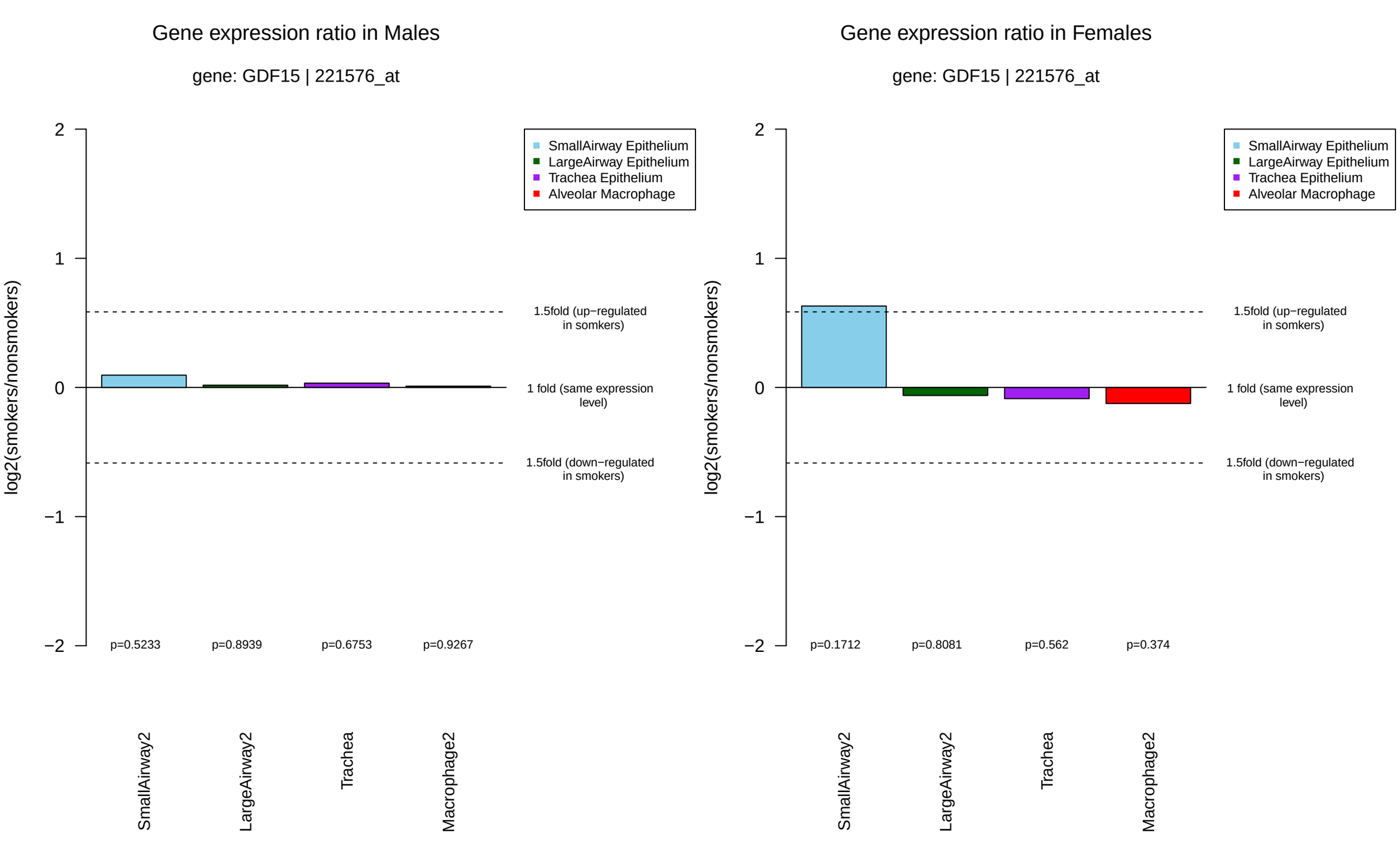
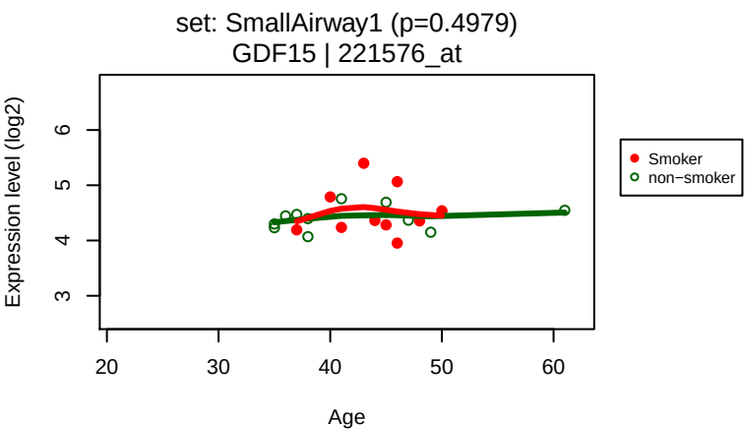
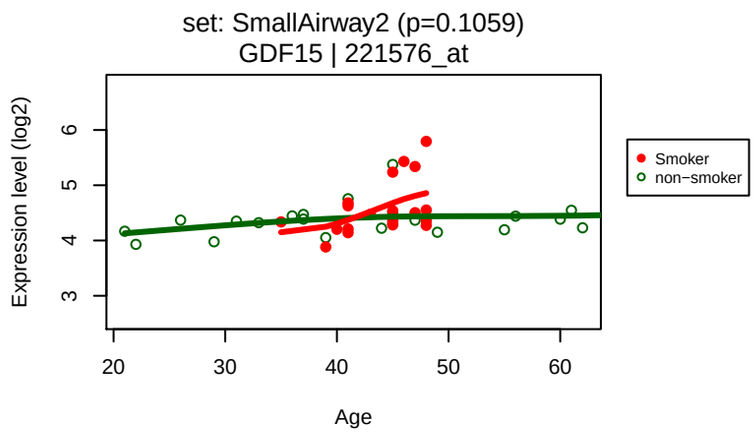
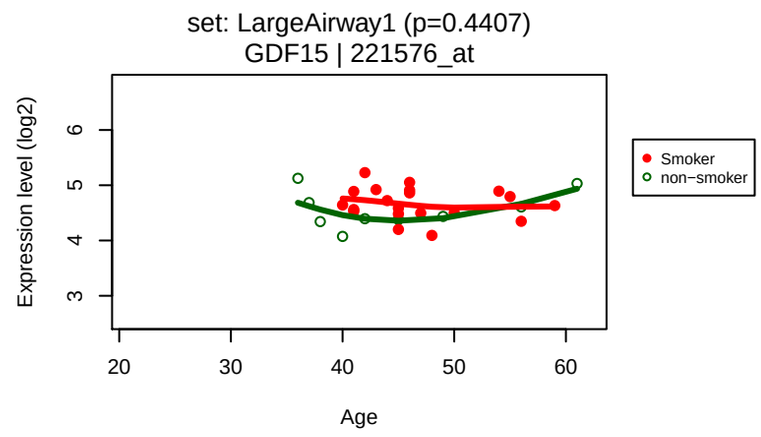

Gene expression of smokers and nonsmokers
gene: GDF15 | 221576_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4979
p=0.1059
p=0.0733
p=0.4407
p=0.8005
p=0.8501
p=0.2342
p=0.6642
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

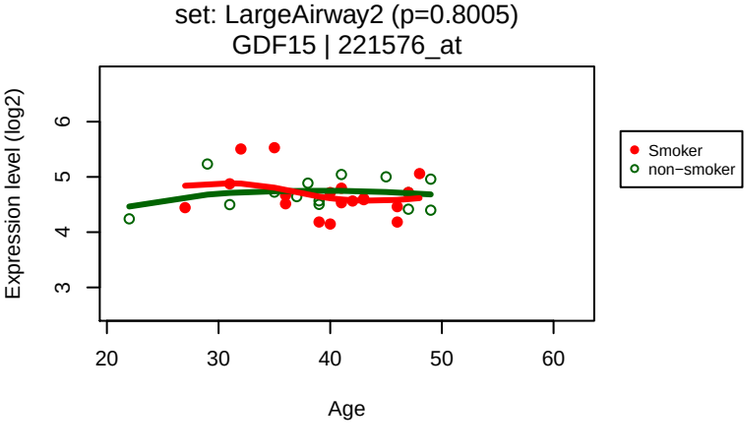

Gene expression ratio (smokers/nonsmokers)
gene: GDF15 | 221576_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4979
p=0.1059
p=0.0733
p=0.4407
p=0.8005
p=0.8501
p=0.2342
p=0.6642
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

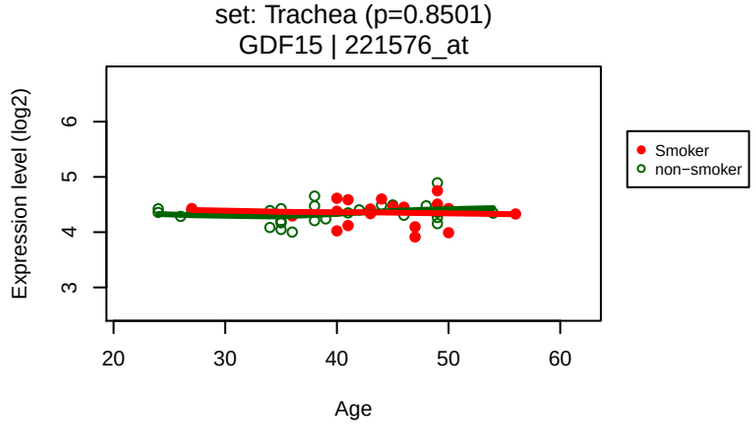

Gene expression ratio in Males
gene: GDF15 | 221576_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5233
p=0.8939
p=0.6753
p=0.9267
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GDF15 | 221576_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1712
p=0.8081
p=0.562
p=0.374
SmallAirway2
LargeAirway2
Trachea
Macrophage2

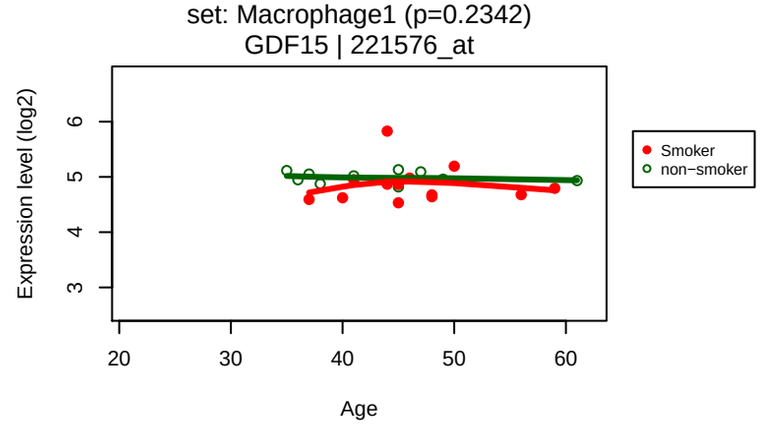

set: SmallAirway1 (p=0.4979)
GDF15 | 221576_at
Expression level (log2)
Age
Smoker
non−smoker

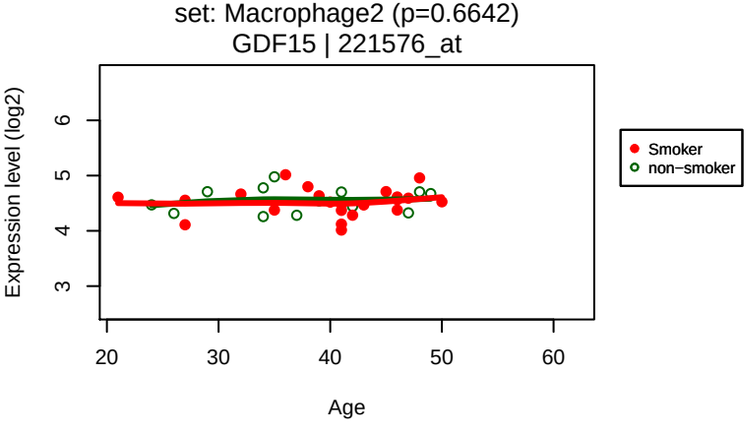

set: SmallAirway2 (p=0.1059)
GDF15 | 221576_at
Expression level (log2)
Age
Smoker
non−smoker

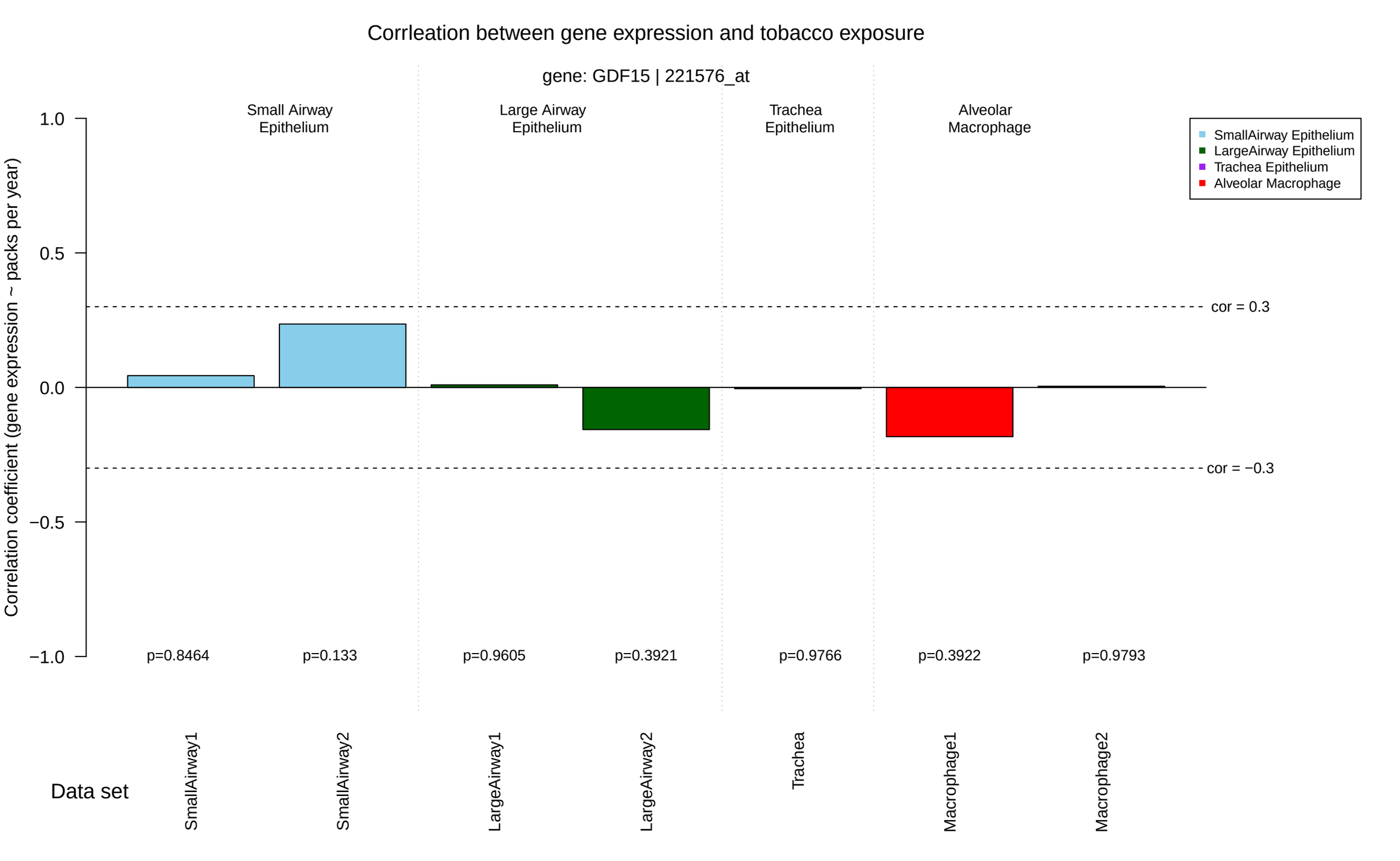

set: LargeAirway1 (p=0.4407)
GDF15 | 221576_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8005)
GDF15 | 221576_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8501)
GDF15 | 221576_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2342)
GDF15 | 221576_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6642)
GDF15 | 221576_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GDF15 | 221576_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8464
p=0.133
p=0.9605
p=0.3921
p=0.9766
p=0.3922
p=0.9793
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2